Lecture 24 (*April 16*) *Reading: Section 8.3*

Viewing: Lecture 28 (after minute 12:00)

The previous class was all about the situation when an $n \times n$ matrix A has distinct eigenvalues. But what if the eigenvalues are repeated, i.e. what if the characteristic polynomial takes the form:

$$p(\lambda) = (d_1 - \lambda)^{r_1}(d_2 - \lambda)^{r_2} \dots (d_s - \lambda)^{r_s} \tag{200}$$

for certain distinct numbers $d_1, \dots, d_s$, and for certain powers $r_1, \dots, r_s$? We think of the eigenvalues of A as being the following sequence of numbers:

$$\underbrace{d_1, \dots, d_1}_{r_1 \text{ times}}, \underbrace{d_2, \dots, d_2}_{r_2 \text{ times}}, \dots, \underbrace{d_s, \dots, d_s}_{r_s \text{ times}} \tag{201}$$

The number r_i is called the **algebraic multiplicity** of the eigenvalue d_i, and it represents the number of times it appears among the roots of the characteristic polynomial. Because the degree of the characteristic polynomial is n, the sum of the algebraic multiplicities of all eigenvalues is:

$$r_1 + \dots + r_s = n \tag{202}$$

What about eigenvectors for each eigenvalue d_i? The set of eigenvectors for any given eigenvalue is a vector subspace of $\mathbb{R}^n$, namely:

$$\Big\{\text{eigenvectors of } A \text{ corresponding to eigenvalue } \lambda\Big\} = N(A - \lambda I)$$

The **geometric multiplicity** of any eigenvalue d_i is defined as the dimension of the corresponding space of eigenvectors, namely $\dim N(A - d_i I)$. It never exceeds the algebraic multiplicity:

$$\boxed{\textbf{geometric multiplicity} \leq \textbf{algebraic multiplicity}} \tag{203}$$

For example, consider the matrices below:

$$A = \begin{bmatrix} d & 0 \\ 0 & d \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} d & 1 \\ 0 & d \end{bmatrix} \tag{204}$$

Both of them have characteristic polynomial $(\lambda - d)^2$, so d is the only eigenvalue, with algebraic multiplicity 2. But the geometric multiplicity differs among the two matrices. In the case of the matrix A, any vector is an eigenvector because $A\boldsymbol{v} = d\boldsymbol{v}$ for all $\boldsymbol{v}$, thus implying that the eigenvalue d has geometric multiplicity 2. But for the matrix B, we have that:

$$\boldsymbol{v} = \begin{bmatrix} x \\ y \end{bmatrix} \text{ is an eigenvector} \quad \Leftrightarrow \quad \begin{bmatrix} d & 1 \\ 0 & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = d \begin{bmatrix} x \\ y \end{bmatrix} \quad \Leftrightarrow \quad y = 0$$

which implies that the subspace of eigenvectors is one-dimensional, hence the eigenvalue d has geometric multiplicity 1. This is strictly smaller than its algebraic multiplicity, which is 2.

Fact 20. *A matrix is diagonalizable if and only if all of its eigenvalues have geometric multiplicity equal to their algebraic multiplicity.*

Indeed, if the geometric multiplicities of all eigenvalues are equal to the corresponding algebraic multiplicities, then (202) implies that the sum of all geometric multiplicities is equal to n. This implies that the subspaces of eigenvectors have dimensions which add up to n, hence we can pick a basis of $\mathbb{R}^n$ consisting only of eigenvectors. This allows us to define the invertible matrix V whose columns are the n chosen eigenvectors, and formula (191) holds.

But what can we say when some of the eigenvalues have geometric multiplicity strictly smaller than their algebraic multiplicity? In this case, the matrix A is no longer similar to a diagonal matrix, as in (191). Instead, it will be similar to a matrix in **Jordan normal form**:

$$A = V \left[\begin{array}{c|c|c|c} J_1 & 0 & \dots & 0 \\ \hline 0 & J_2 & \dots & 0 \\ \hline \vdots & \vdots & \ddots & \vdots \\ \hline 0 & 0 & \dots & J_t \end{array}\right] V^{-1} \tag{205}$$

where each J_i is a **Jordan block**, i.e. a matrix of the form:

$$\begin{bmatrix} \lambda & 1 & 0 & 0 & \dots & 0 \\ 0 & \lambda & 1 & 0 & \dots & 0 \\ 0 & 0 & \lambda & 1 & \dots & 0 \\ 0 & 0 & 0 & \lambda & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & \lambda \end{bmatrix} \tag{206}$$

where λ is among the eigenvalues of A. The total number of d_i's that appear on the diagonal of the block matrix in (205) is equal to r_i, the algebraic multiplicity of the eigenvalue d_i.

There are two things one needs to know in order to set up the Jordan normal form (205): the specific sizes of the Jordan blocks $J_1, \dots, J_t$ and the matrix V. There is a general prescription for figuring out the sizes of the Jordan blocks, but it goes beyond the scope of our course. As for the matrix V, we know from the diagonalizable case that its columns have something to do with the nullspaces $N(A - d_1 I), \dots, N(A - d_s I)$. However, in the situation at hand (when the geometric multiplicities are smaller than the algebraic multiplicities), the dimensions of these nullspaces do not add up to n, so they will not contain n linearly independent vectors. So instead, look at:

$$N\Big((A - d_1 I)^{r_1}\Big), \dots, N\Big((A - d_s I)^{r_s}\Big) \tag{207}$$

(where $r_1, \dots, r_s$ are the algebraic multiplicities of the eigenvalues) and we claim that these subspaces are both linearly independent, and span the entire $\mathbb{R}^n$. Then the subspaces (207) will contain n linearly independent vectors, and these will be the columns of the matrix V of (205).

Let's present this in more detail in the case when the algebraic multiplicities are all either 1 or 2. Consider the matrix:

$$A = \begin{bmatrix} 10 & -3 & 7 \\ 27 & -10 & 18 \\ 5 & -3 & 3 \end{bmatrix}$$

and we want to compute its Jordan normal form: i.e. a matrix V and Jordan blocks $J_1, \dots, J_t$ such

that formula (205) holds. First of all, let's compute the characteristic polynomial:

$$p(\lambda) = \det(A - \lambda I) = \det \begin{bmatrix} 10-\lambda & -3 & 7 \\ 27 & -10-\lambda & 18 \\ 5 & -3 & 3-\lambda \end{bmatrix}$$

Let us compute the determinant by cofactor expansion along the last row:

$$p(\lambda) = 5 \det \begin{bmatrix} -3 & 7 \\ -10-\lambda & 18 \end{bmatrix} - (-3) \det \begin{bmatrix} 10-\lambda & 7 \\ 27 & 18 \end{bmatrix} + (3-\lambda) \det \begin{bmatrix} 10-\lambda & -3 \\ 27 & -10-\lambda \end{bmatrix} =$$

$$= 5(-54 + 70 + 7\lambda) + 3(180 - 18\lambda - 189) + (3-\lambda)(\lambda^2 - 100 + 81) = -\lambda^3 + 3\lambda^2 - 4$$

It's not easy to find the roots of a general cubic polynomial. But when you're out of ideas, just try plugging in some simple numbers for λ to see if $p(\lambda) = 0$. For example, trying $\lambda = -1$ shows that $p(-1) = 0$, which implies that $p(\lambda)$ is divisible by $\lambda + 1$. Then polynomial long division shows that:

$$p(\lambda) = (\lambda+1)(-\lambda^2 + 4\lambda - 4) = -(\lambda+1)(\lambda-2)^2$$

So the eigenvalues are $d_1 = -1$ with algebraic multiplicity one, and $d_2 = 2$ with algebraic multiplicity two. Now for the subspaces of eigenvectors. We have:

$$A + I = \begin{bmatrix} 11 & -3 & 7 \\ 27 & -9 & 18 \\ 5 & -3 & 4 \end{bmatrix} \overset{\text{RREF}}{\rightsquigarrow} \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \quad \text{hence} \quad N(A+I) = N\left(\begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \right)$$

The matrix on the right has pivot columns 1 and 2, and free column 3, so its null-space is one dimensional and spanned by the vector:

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad \text{such that} \quad \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = 0 \qquad \Rightarrow x = -\frac{1}{2} \text{ and } y = \frac{1}{2}$$

Therefore, the eigenvalue $d_1 = -1$ also has geometric multiplicity one, and an eigenvector is:

$$\boldsymbol{v}_1 = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix}$$

As for the other eigenvalue, we have:

$$A - 2I = \begin{bmatrix} 8 & -3 & 7 \\ 27 & -12 & 18 \\ 5 & -3 & 1 \end{bmatrix} \overset{\text{RREF}}{\rightsquigarrow} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{hence} \quad N(A-2I) = N\left(\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \right)$$

The matrix on the right has pivot columns 1 and 2, and free column 3, so its null-space is one dimensional and spanned by the vector:

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad \text{such that} \quad \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = 0 \qquad \Rightarrow x = -2 \text{ and } y = -3$$

Therefore, the eigenvalue $d_2 = 2$ has geometric multiplicity one, and an eigenvector is:

$$\boldsymbol{v}_2 = \begin{bmatrix} -2 \\ -3 \\ 1 \end{bmatrix}$$

So we are in the situation when geometric multiplicity is strictly smaller than algebraic multiplicity, hence the subspaces of eigenvectors are not enough to produce a basis of $\mathbb{R}^3$. In this case, (207) tells us to replace the nullspace of $A - 2I$ by the nullspace of $(A - 2I)^2$, namely:

$$(A-2I)^2 = \begin{bmatrix} 18 & -9 & 9 \\ -18 & 9 & -9 \\ -36 & 18 & -18 \end{bmatrix} \overset{\text{RREF}}{\rightsquigarrow} \begin{bmatrix} 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{hence} \quad N((A-2I)^2) = N\left(\begin{bmatrix} 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\right)$$

The matrix on the right has pivot column 1 and free columns 2 and 3, so its nullspace is two dimensional and consists of vectors:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \qquad \text{such that} \quad 2x - y + z = 0$$

One of these vectors is $\boldsymbol{v}_2$, but we can take any other such vector, let's say $\boldsymbol{v}_3 = \begin{bmatrix} a \\ 2a \\ 0 \end{bmatrix}$ (there will be a benefit in making the last entry 0, as we will soon see) for any a. Therefore, the matrix:

$$V = \left[\begin{array}{c|c|c} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 \end{array}\right] = \begin{bmatrix} -\frac{1}{2} & -2 & a \\ \frac{1}{2} & -3 & 2a \\ 1 & 1 & 0 \end{bmatrix}$$

is invertible. The constant $a \neq 0$ is yours to choose freely, but if you want to have the equality:

$$A = V \left[\begin{array}{c|cc} -1 & 0 & 0 \\ \hline 0 & 2 & 1 \\ 0 & 0 & 2 \end{array}\right] V^{-1}$$

then you need to choose $a = -1$ (if you were to use any other value for a, then you would just need to replace the entry 1 in the 2×2 Jordan block above by the entry $-a$, which is not too bad).

Remark. *As I mentioned, I did not give you a complete recipe for how to set up the Jordan blocks in* (205)*. But as you were able to see from the example above, if the Jordan blocks corresponding to an eigenvalue λ are located on columns $i, \ldots, j$, then the columns $i, \ldots, j$ of the matrix V need to be filled with a basis of $N((A-\lambda I)^r)$, where r is the algebraic multiplicity of the eigenvalue λ.*